
Introductory Statistics for Econometrics

Unit 1. Probability Foundations

Sample spaces, events, axioms of probability,
counting rules, conditional probability and independence

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Learning outcomes. After this unit, the student should be able to (i) describe a random experiment in terms of a sample space and events; (ii) operate with events using set theory; (iii) state and use the axioms of probability; (iv) apply elementary counting rules; (v) compute and interpret conditional probabilities; and (vi) recognise (conditional) independence and use the Law of Total Probability.

Source material: condensed from MIT OCW 14.30 (Spring 2009), K. Menzel, Lecture Notes 1–3.

1 Random experiments, sample space and events

Probability theory provides a mathematical language to describe situations in which we do not have perfect knowledge of all relevant factors. In economics and finance, virtually every interesting question — from asset pricing to treatment evaluation — is posed in this language.

Definition 1.1 (Random experiment). A *random experiment* is any procedure which (a) can, at least theoretically, be repeated arbitrarily often under identical conditions, and (b) has a well-defined set of possible outcomes.

Definition 1.2 (Sample space). The *sample space* $\mathcal{S}$ is the collection of all possible outcomes of a random experiment.

Definition 1.3 (Event). An *event* A is any collection of outcomes, $A \subseteq \mathcal{S}$. If the realised outcome is a member of A , we say that A *has occurred*.

Definition 1.4 (Empty set). The *empty set* $\emptyset$ is a set without outcomes, $\emptyset = \{\}$. Note $\emptyset \neq \{0\}$ and $\emptyset \neq \{\emptyset\}$. Second inequality comes from the fact that $\{\emptyset\}$ is a set **with one element**, which is the empty set. Therefore, by definition of the emptyset, $\emptyset \neq \{\emptyset\}$

Example 1.5. Roll one fair die. The sample space is $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$. The event “the outcome is even” is $A = \{2, 4, 6\}$.

2 Set operations on events

Because events are sets, the algebra of events is the algebra of sets. Let $A, B \subseteq \mathcal{S}$.

Inclusion. $A \subseteq B$ iff every outcome of A is also in B : “ A implies B ”. In particular $\emptyset \subseteq A \subseteq \mathcal{S}$.

Union. $A \cup B = \{s \in \mathcal{S} : s \in A \text{ or } s \in B\}$ (“ A or B ”).

Intersection. $A \cap B = \{s \in \mathcal{S} : s \in A \text{ and } s \in B\}$ (“ A and B ”). When unambiguous we also write $A \cap B \equiv AB$.

Complement. $A^c = \{s \in \mathcal{S} : s \notin A\}$ (“not A ”).

The key algebraic identities are:

$$\begin{aligned} \text{commutative: } & A \cup B = B \cup A, & A \cap B &= B \cap A, \\ \text{associative: } & (A \cup B) \cup C = A \cup (B \cup C), & (A \cap B) \cap C &= A \cap (B \cap C), \\ \text{distributive: } & A \cap (B \cup C) = (A \cap B) \cup (A \cap C), & A \cup (B \cap C) &= (A \cup B) \cap (A \cup C), \\ \text{De Morgan: } & (A \cup B)^c = A^c \cap B^c, & (A \cap B)^c &= A^c \cup B^c. \end{aligned}$$

Definition 2.1 (Disjoint events, partition). A and B are *disjoint* (mutually exclusive) if $A \cap B = \emptyset$. A countable collection $\{A_i\}_{i \geq 1}$ is a *partition* of $\mathcal{S}$ if its members are pairwise disjoint and $\bigcup_i A_i = \mathcal{S}$.

3 Axioms of probability and consequences

A probability is a function $\mathbb{P} : \mathcal{A} \rightarrow \mathbb{R}$ that assigns a number to each event in a suitable family $\mathcal{A}$ (a σ -algebra; the technicalities are not essential for this course).

Definition 3.1 (Kolmogorov’s axioms). A function $\mathbb{P}$ is a *probability* if it satisfies:

(P1) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{A}$;

(P2) $\mathbb{P}(\mathcal{S}) = 1$;

(P3) For any sequence of *disjoint* events $A_1, A_2, \dots$, $\mathbb{P}\left(\bigcup_{i \geq 1} A_i\right) = \sum_{i \geq 1} \mathbb{P}(A_i)$.

The axioms are *minimal* requirements: they do not pin down a unique probability. Any function satisfying (P1)–(P3) is a valid probability; whether it accurately models a particular phenomenon is a separate empirical question.

Proposition 3.2 (Basic properties). *For any events $A, B \in \mathcal{A}$:*

(i) $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$;

(ii) $\mathbb{P}(\emptyset) = 0$;

(iii) $A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B)$ (monotonicity);

(iv) $0 \leq \mathbb{P}(A) \leq 1$;

(v) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$ (inclusion–exclusion).

Proof of (i). Since $\mathcal{S} = A \cup A^c$ and $A \cap A^c = \emptyset$, axioms (P2) and (P3) give $1 = \mathbb{P}(\mathcal{S}) = \mathbb{P}(A) + \mathbb{P}(A^c)$. The remaining items follow by similar partition arguments; see *Exercises*. $\square$

Equally likely outcomes. When $\mathcal{S}$ is finite and all outcomes are *ex ante* symmetric (“fair” device), the probability of any event reduces to a counting problem:

$$\mathbb{P}(A) = \frac{\#A}{\#\mathcal{S}}. \quad (1)$$

The remainder of this section gives tools to compute the numerator and denominator in (1).

4 Counting rules

(1) Multiplication rule. If a procedure has two stages, the first with m possible outcomes and the second with n possible outcomes (regardless of the first), the joint procedure has $m \cdot n$ outcomes. By induction, k stages with $n_1, \dots, n_k$ outcomes give $\prod_{i=1}^k n_i$ outcomes.

(2) Sampling with replacement. Drawing n items from a population of size N , where each draw is independent of the previous ones, gives N^n ordered outcomes.

(3) Permutations (sampling without replacement, ordered). Drawing n items from N without replacement, where order matters:

$$P_{N,n} = N(N-1)\cdots(N-n+1) = \frac{N!}{(N-n)!}.$$

(4) Combinations (sampling without replacement, unordered). Drawing n items from N where order does *not* matter:

$$\binom{N}{n} = \frac{N!}{(N-n)!n!}.$$

Example 4.1 (Birthday paradox). In a group of n people, the probability that at least two share a birthday is, assuming uniform birthdays,

$$\mathbb{P}(\text{coincidence}) = 1 - \frac{365!}{(365-n)!365^n}.$$

For $n = 23$ this already exceeds 50%. The lesson — central to econometric intuition about “data mining” — is that the number of *potential* coincidences grows much faster than the number of people.

5 Conditional probability

Probabilities are revised when new information arrives. Given that an event B with $\mathbb{P}(B) > 0$ has occurred, the relevant sample space shrinks to B and probabilities must be re-normalised.

Definition 5.1 (Conditional probability). For events A, B with $\mathbb{P}(B) > 0$,

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}. \quad (2)$$

Equation (2) immediately yields the *multiplication rule for probabilities*,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A | B) \mathbb{P}(B) = \mathbb{P}(B | A) \mathbb{P}(A). \quad (3)$$

Example 5.2 (Fair die). Roll a fair die. Conditional on the outcome being even, $B = \{2, 4, 6\}$, the probability of rolling a 6 is $\mathbb{P}(\{6\} | B) = \mathbb{P}(\{6\})/\mathbb{P}(B) = (1/6)/(1/2) = 1/3$.

Remark 5.3. Conditional probabilities will reappear throughout the course: the conditional distribution of one random variable given another (Unit 3), the conditional expectation function (Unit 6), and Bayesian updating (Unit 2). They are also the technical core of econometric “identification” arguments.

6 Independence

Two events are *independent* when knowing that one occurred provides no information about the other.

Definition 6.1 (Independence). Events A and B are independent, written $A \perp B$, if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) \mathbb{P}(B).$$

Equivalently, when $\mathbb{P}(B) > 0$, $\mathbb{P}(A | B) = \mathbb{P}(A)$.

A finite (or countable) collection $\{A_1, \dots, A_n\}$ is *(mutually) independent* if for every subcollection $\{A_{i_1}, \dots, A_{i_k}\}$ one has $\mathbb{P}(A_{i_1} \cap \dots \cap A_{i_k}) = \prod_{j=1}^k \mathbb{P}(A_{i_j})$. Pairwise independence does *not* imply mutual independence.

Proposition 6.2. *If A and B are independent, so are A and B^c , A^c and B , and A^c and B^c .*

Definition 6.3 (Conditional independence). Events A and B are independent *conditional on* C (with $\mathbb{P}(C) > 0$) if $\mathbb{P}(A \cap B \mid C) = \mathbb{P}(A \mid C)\mathbb{P}(B \mid C)$.

Warning. Unconditional independence does *not* imply conditional independence, nor vice versa. Whether two events are independent depends crucially on the conditioning information. This subtlety is the econometric content of phrases such as “controlling for X ”.

7 Law of Total Probability

The Law of Total Probability decomposes an unconditional probability into a weighted average of conditional probabilities. It is the discrete analogue of the law of iterated expectations to be introduced in Unit 6.

Theorem 7.1 (Law of Total Probability). *Let $\{B_1, \dots, B_n\}$ be a partition of $\mathcal{S}$ with $\mathbb{P}(B_i) > 0$. Then, for any event A ,*

$$\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \mid B_i) \mathbb{P}(B_i). \quad (4)$$

Proof. Since $\{B_i\}$ partitions $\mathcal{S}$, the sets $\{A \cap B_i\}$ partition A . By (P3), $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$. $\square$

Example 7.2 (Composition effect / Simpson’s paradox). Suppose mortality rates conditional on procedure type ($B_i \in \{\text{single, double, triple, quadruple bypass}\}$) are systematically lower for experienced than for inexperienced surgeons, but experienced surgeons are also assigned the harder cases. The unconditional (“marginal”) mortality of experienced surgeons can then be *higher* — even though they perform better in every category. Equation (4) reconciles the apparent paradox: the weights $\mathbb{P}(B_i)$ differ between groups. Untangling such composition effects is one of the central tasks of regression analysis.

Summary

- A random experiment is described by a sample space $\mathcal{S}$, a family of events $\mathcal{A}$, and a probability $\mathbb{P}$.
- Set operations ($\cup, \cap, ^c$) on events obey commutative, associative, distributive and De Morgan laws.
- Kolmogorov’s axioms (P1)–(P3) imply: $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$, monotonicity, and inclusion–exclusion.
- For finite, equally likely outcomes, $\mathbb{P}(A) = \#A / \#\mathcal{S}$, computed via multiplication, permutations, and combinations.
- $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B) / \mathbb{P}(B)$; $A \perp B \Leftrightarrow \mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.
- Law of Total Probability: $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i) \mathbb{P}(B_i)$ for a partition $\{B_i\}$.

Suggested exercises

1. Prove parts (ii)–(v) of Proposition 3.2.
2. A fair coin is tossed three times. Write down $\mathcal{S}$. Compute the probability of (a) at least two heads, (b) exactly two heads, and (c) the second toss being a head given that there are at least two heads.
3. Show that if $\mathbb{P}(A) = 0$ or $\mathbb{P}(A) = 1$, then A is independent of every event $B \in \mathcal{A}$.
4. (Counterexample to pairwise $\Rightarrow$ mutual independence.) Take $\mathcal{S} = \{s_1, s_2, s_3, s_4\}$ uniform. Let $A = \{s_1, s_2\}$, $B = \{s_1, s_3\}$, $C = \{s_1, s_4\}$. Verify pairwise independence and show $\mathbb{P}(A \cap B \cap C) \neq \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$.
5. (Composition effect.) Construct a numerical example with two groups and two categories in which group 1 has higher conditional probabilities in every category but lower marginal probability than group 2.